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Assignment - 04  
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Solution - A

$$x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$$

1. start with first two congruences.

Let,  $x = 3a$ , substitute into the second congruence

$$3a + 1 \equiv 2 \pmod{5}$$

$$\Rightarrow 3a \equiv 1 \pmod{5}$$

The inverse of 3 (mod 5) is 2, so

$$a \equiv 2 \pmod{5}$$

$$\Rightarrow a = 5b + 2$$

$$\text{Then } x = 3(5b + 2) + 1 = 15b + 7$$

2. now all three congruence

$$\text{Since } 15 \equiv 1 \pmod{7} \text{ and } 7 \equiv 0 \pmod{7}$$

$$b \equiv 3 \pmod{7} \Rightarrow b = 7c + 3$$

$$x \equiv 52 \pmod{15} \Rightarrow 52$$

Solution  $\rightarrow B$

$$x \equiv 5 \pmod{8}, x \equiv 11 \pmod{29}, x \equiv 15 \pmod{31}$$

Total modulus:

$$M = 8 \times 29 \times 31 = 7192$$

$$M_1 = 7192/8 = 899, 899 \equiv 3 \pmod{8}, 3^{-1} \equiv 3 \pmod{8}$$

$$M_2 = 7192/29 = 248, 248 \equiv 16 \pmod{29}, 16^{-1} \equiv 20 \pmod{29}$$

$$M_3 = 7192/31 = 232, 232 \equiv 15 \pmod{31}, 15^{-1} \equiv 20 \pmod{31}$$

$$\text{Sum } x \equiv 5 \cdot 899 \cdot 3 + 11 \cdot 248 \cdot 20 + 15 \cdot 232 \cdot 20 = 134245 \pmod{7192}$$

$$\text{Reduce } 134245 \equiv 4789 \pmod{7192}$$

$$x \equiv 4789 \pmod{7192} \text{ Ans.}$$

Solution  $\rightarrow C$

$$x \equiv 5 \pmod{8}, x \equiv 4 \pmod{11}, x \equiv 3 \pmod{17}$$

$$M = 8 \cdot 11 \cdot 17 = 1496$$

$$M_1 = 1496/8 = 187, 187 \equiv 3 \pmod{8}, 3^{-1} \equiv 3 \pmod{8}$$

$$M_2 = 1496/11 = 136, 136 \equiv 4 \pmod{11}, 4^{-1} \equiv 3 \pmod{11}$$

$$M_3 = 1496/17 = 88, 88 \equiv 3 \pmod{17}, 3^{-1} \equiv 6 \pmod{17}$$

$$\text{Sum } \equiv 5 \cdot 187 \cdot 3 + 4 \cdot 136 \cdot 3 + 3 \cdot 88 \cdot 6 = 6021 \pmod{1496}$$

$$\text{Reduce } 6021 \equiv 37 \pmod{1496}$$

$$x \equiv 37 \pmod{1496} \text{ Ans.}$$